

Did it actually work? Evaluating an innovative solution.

Building a solution proves you can. Evaluating it answers the harder question: was it worth it? This module is about judging a finished solution on two separate scales — effectiveness (how well it does the job) and efficiency (how little it wastes doing it) — using criteria you wrote before you built, and evidence you

What this key knowledge covers

This summary distils everything you need for **KK14 — Did it actually work? Evaluating an innovative solution.** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

The dot point, in plain English

Unit 2 Outcome 1 walks the whole **problem-solving methodology** — Analysis → Design → Development → Evaluation — to build one innovative solution. This dot point is the **finish line**. After your solution has been out in the world being used, you step back and judge it. Two things are examinable here, and the exam loves to test whether you can keep them apart:

The **questions you ask** to decide if the solution succeeded — written as specific, measurable statements during *design*, covering both efficiency and effectiveness.

The **methods you use** to gather evidence against each criterion after real use — interviews, surveys, observation, and analysing the solution's own logged data.

Back where the chapter began

This chapter opened with FloatMate — the wearable rip-current alert wristband that **Jade**, a Year 11 Nippers volunteer, dreamed up at her surf club. We watched the idea form (KK01), get analysed and designed, then developed. Now it's the end of summer. FloatMate wristbands have been worn by junior swimmers through a **full season of patrols**. The club committee wants one thing from Jade: *tell us, honestly, whether this was a good solution*.

That word — *honestly* — is the whole game. Jade can't just say "I worked really hard and it looks great." She has to **measure it against the criteria she wrote during design**, using evidence she collected once kids actually wore it in the surf. That's evaluation.

- Detect when a wearer drifts into a rip
- Alert that child clearly and immediately

- Cut the number of dangerous drifts and rescues
- Be comfortable enough that kids keep wearing it
- A battery that lasts a whole patrol shift
- An alert that fires within a second or two
- A unit price the club can actually afford
- Setup that doesn't eat a volunteer's whole morning

The left column is about effectiveness — hitting the goal. The right column is about efficiency — not wasting time, money, battery or effort getting there. Hold that split in your head; the rest of the module sharpens it.

The distinction examiners bait you with

Students lose easy marks every year by treating **testing** and **evaluation** as the same thing. They are different jobs, at different times, answering different questions.

- Question: *does it work the way I built it to?*
- Happens while you're still coding the solution
- Uses test data and test cases (expected vs actual)
- Finds and fixes faults before release
- You run it yourself, in the workshop
- Question: *was it a success for the people who use it?*
- Happens weeks/months after release, in the real world
- Measures against evaluation criteria using real evidence
- Judges whether it met the original need
- You gather data from users, not just yourself

When each thing happens on the methodology timeline

Slide through the stages. Notice where the criteria are written, where testing lives, and how far away evaluation sits.

Effectiveness — how well it does the job

Think of a chef cooking dinner for a hungry crowd. **Effectiveness** is whether everyone leaves full and the food tastes great — the goal got hit. It says nothing yet about how long it took or how much was wasted. A meal can be wonderfully effective and a total disaster on resources.

Common effectiveness measures

Right outputs, no errors. FloatMate: does it flag a real rip and not cry wolf?

Does it do *all* of the job, not just part? Every required feature present.

Can a young Nipper understand the alert and act on it instantly?

Works for the range of users it's meant for, not just the easy cases.

Solves the *actual* need — fewer dangerous drifts — not a tidier near-miss.

Delivers the message or outcome the functional requirements demanded.

Quick test: if a measure describes *how good the result is for the user*, it's effectiveness.

Efficiency — how little it wastes

Back to the chef. **Efficiency** is whether that great meal took 30 minutes or 5 hours, used half the pantry or just what was needed, ran the oven all day or barely at all. Same delicious result, wildly different efficiency. This is exactly why you must measure the two separately.

Common efficiency measures

Speed to produce a result. FloatMate: seconds from rip-entry to alert.

Money to build, buy or run. The price per wristband the club must pay.

Human work to set up and maintain — minutes of volunteer time each patrol.

Battery, data, processing, materials — what the solution consumes to run.

Output per unit of input — more swimmers covered per charged band.

How quickly the system computes/handles its workload under load.

Why a solution needs both scales

Effective and efficient are independent axes, so a solution lands in one of four quadrants.

Picture a delivery drone dropping a parcel on a target pad. **Effectiveness** is how close the parcel lands to the centre of the pad. **Efficiency** is how much battery the drone had left when it got there. Drive both sliders and watch the verdict change.

Notice the bottom-right: a drone that nails the target but lands on empty is still a *problem* — it can't do the next run. Effectiveness without efficiency isn't "good enough", and a cheap, fast solution that misses the target is worthless. Evaluation reports on both, separately, then weighs them.

What makes a criterion examinable

An **evaluation criterion** is a single, specific, measurable statement (often phrased as a question) you use to judge success. A good set of criteria shares four traits:

You could collect a number or a clear yes/no. "FloatMate is good" fails. "What % of real rip events triggered an alert?" works.

Set *before* building, so the result can't be bent to flatter them. They flow from the solution requirements.

Each criterion checks something the solution actually needed to do or be — not a random nice-to-have.

Some criteria probe effectiveness, others efficiency. A set that's all one kind is incomplete.

Anatomy of a strong criterion

Criterion: Does FloatMate alert the wearer for at least 95% of genuine rip events?

Measure / target: count alerts vs logged rip events over the season; target $\geq 95\%$.

Every criterion needs a way to measure it

A criterion is only useful if you can gather evidence for it. So each one is paired with a **technique**. Prefer an **objective** method (a hard number anyone would read the same way) and fall back to a **subjective** method (people's opinions) only when no number exists.

* Analysing the band's **logged data**: alerts fired, response times, battery drain

- **Timing** the alert delay across many events
- **Comparing incident reports** before vs after the trial
- Measuring **cost** per unit against the budget
- **Interviews** with lifesavers and Nippers
- **Surveys / questionnaires** on comfort and clarity
- **Observation** of how kids respond to an alert on patrol
- User **ratings** of how easy the alert was to understand

"Was the alert clear?" can't be read off a sensor — that's a job for a survey or interview (subjective). "How fast did it fire?" is a stopwatch job (objective). Match the method to what the criterion is actually asking.

Match each criterion to a sensible method

Pick the best evaluation technique for each. Choices update live.

Effectiveness or efficiency?

Drag each measure into the right bucket — or tap a chip then tap a bucket. Then check your answers.

From vague to measurable

These criteria would lose marks. Tap each to reveal a stronger version and why it's better.

✓ "What percentage of genuine rip events did FloatMate alert on across the season? (target $\geq$ 95%)"

"Reliable" can't be measured. The fix names a number, a source of evidence and a target — and it's clearly an effectiveness criterion.

✓ Split into three: alert delay (seconds), unit cost (\$ vs budget), and alert-clarity rating — each measured separately.

One criterion should test *one* thing. Bundling three makes it impossible to say which part passed or failed.

✓ "After a season of use, did logged rescue/near-miss counts fall compared with the previous season?"

Testing $\neq$ evaluation. A real evaluation criterion looks at **outcomes after real use**, with evidence gathered from the world, not the workshop.

✓ "How much volunteer time did set-up take per patrol?" (an efficiency measure) — and keep "liked the colour" as a separate effectiveness/appeal note.

Five questions · 18 marks · mark yourself

Have a real go first. Then reveal the weak answer, the strong answer and the marking guide, and award yourself a score — it feeds the dock in the corner.

Everything KK14, on a postcard

Effectiveness = how well it does the job (accuracy, completeness, clarity, meets the need).

Efficiency = how little it wastes (time, cost, effort, resources).

Testing = during development, "does it work?". Evaluation = after a period of real use, "was it a success against criteria?".

Specific, measurable, from the requirements, covering both scales — written before building so the verdict stays honest.

Objective first (logs, timing, comparing records, costs); subjective fallback (interviews, surveys, observation).

On-target but flat battery = effective, not efficient. Cheap and fast but off-pad = efficient, not effective. Report both, then weigh.

- * Anchor evaluation to "after real use".
- * Give every criterion a measure.
- * Always cover both scales + a judgement.

Common traps & misconceptions

Calling testing "evaluation"

1 Testing checks it *works* during the build; evaluation judges *success* after real use. Use the word "after a period of use" to anchor it. **Fix:** name the timing and the evidence source.

Swapping efficiency and effectiveness

2 Battery/time/cost/effort = efficiency. Accuracy/clarity/completeness/meets-the-need = effectiveness. **Fix:** ask "is this about the *result* or the *resources*?"

Criteria with no measure

3 "It should be good/fast/nice" earns nothing. **Fix:** give every criterion a number, target or clear yes/no and a way to gather it.

Inventing criteria after the result

4 Criteria are written at **design**, before building. **Fix:** if a "describe" question asks when criteria are created, say the design stage — and explain it keeps the evaluation honest.

Evaluating only one scale

5 Saying it's effective but never mentioning efficiency (or vice versa) loses half the question. **Fix:** always cover both, even briefly.

Listing a method without saying what it measures

6 "Do a survey" isn't enough. **Fix:** tie the technique to the criterion — *survey users to measure how clear the alert was*.

Confusing "efficient" with "effective" in the verdict

7 A solution can hit its goal yet waste resources. **Fix:** report each judgement separately, then give an overall weighing.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Distinguish · 2 marks

Between *testing* and *evaluation* of an innovative solution.

✗ A weak answer

— would score ~0–1 "Testing and evaluation both check the solution is okay and that it works properly." No actual distinction — it says they're the same. No mention of *when* each happens or what each judges.

✓ A strong answer

— full marks "Testing happens **during development** to check the solution functions correctly, using test data and expected-vs-actual results. Evaluation happens **after a period of real use** and judges whether the solution actually met the user's needs, measured against pre-set criteria using real evidence."

How the marks are awarded

- 1 mark: testing = during development, checks it works/functions correctly.
- 1 mark: evaluation = after a period of use, judges success/meeting needs against criteria.
- Must show a clear contrast in *timing* and *purpose* for both marks.

Q2. Evaluate · 3 marks

Jade must evaluate FloatMate. Write **one effectiveness criterion** and **one efficiency criterion**, and for each name a technique she could use to gather evidence.

✗ A weak answer

— would score ~1 "Effectiveness: it should work well. Efficiency: it should be fast. She can test it." Neither criterion is measurable, and "test it" isn't an evaluation technique. The efficiency/effectiveness labels are barely justified.

✓ A strong answer

— full marks "**Effectiveness criterion:** What percentage of genuine rip events triggered an alert? — gathered by *analysing the band's logged event data* against lifesaver incident reports. **Efficiency criterion:** How many seconds elapsed between rip-entry and the alert firing? — gathered by *timing logged alert delays* across the season."

How the marks are awarded

- 1 mark: a measurable *effectiveness* criterion (about the result/goal).
- 1 mark: a measurable *efficiency* criterion (about time/cost/effort/resources).
- 1 mark: a sensible technique correctly paired to *at least one* criterion.

Q3. Explain · 3 marks

Why evaluation criteria should be created during the *design* stage rather than after the solution is finished.

✗ A weak answer

— would score ~1 "Because you need them later to check the solution, so it's good to have them ready early." States the timing but gives no real reason — misses the honesty/bias point that earns the higher marks.

✓ A strong answer

— full marks "Writing criteria at the design stage means they come straight from the original solution requirements, so they test what the solution was actually meant to do. It also keeps the evaluation honest: if criteria were written after seeing the results, you could unconsciously choose ones the solution happens to pass, biasing the judgement. Setting them early gives a fixed, fair benchmark to measure against."

How the marks are awarded

- 1 mark: criteria derive from the solution requirements / what it was meant to do.
- 1 mark: avoids bias — can't bend criteria to flatter the result.
- 1 mark: provides a fixed, fair benchmark agreed before building.

Q4. Evaluate · 4 marks

Transfer: A school library installs **ShelfScan**, a self-checkout kiosk students built. After a term, evaluate ShelfScan in terms of both efficiency and effectiveness, naming a measure for each.

✗ A weak answer

— would score ~1–2 "ShelfScan is effective because it's quick and efficient because students like it. Overall it's good." "Quick" is wrongly filed under effectiveness and "students like it" under efficiency — both swapped. No measures, no real judgement.

✓ A strong answer

— full marks "**Effectiveness:** measure the proportion of loans completed without staff help — comparing borrowing logs and observing students shows whether ShelfScan actually does the borrowing job for the people using it. **Efficiency:** measure the average time per checkout and the staff-hours saved at the desk — fewer seconds and fewer staff minutes mean better use of resources. **Judgement:** if most loans succeed unaided (effective) and average checkout time and staffing both fall (efficient), ShelfScan is a success on both scales; if loans frequently fail, it's efficient but not effective and would need rework."

How the marks are awarded

- 1 mark: a correct *effectiveness* measure (about doing the job/result).
- 1 mark: a correct *efficiency* measure (time/cost/effort/resources).
- 1 mark: evidence/method tied to the measures (logs, observation, timing).
- 1 mark: an overall judgement that weighs both scales.